Band-1 Legendre: CONDITIONAL Probe under RH / Zero-Free (Collapse ~2.25)

Christopher Baird + ZoraASI Band-1 dash (executor)

2026-09-11

Band-1 Legendre Conjecture — CONDITIONAL Probe (Collapse ≈ 2.25)

Honesty stamp. This note is a **CONDITIONAL** literature + finite-box probe. It does **not** prove Legendre's conjecture unconditionally. It does **not** claim that the Riemann Hypothesis implies Legendre for consecutive squares (that implication is still open). It does **not** claim a Clay-millennial solution or a completed ToE. **GATE-04** is locked out of scope.

> **Label: CONDITIONAL.** All RH / zero-free statements below are labeled as such. Unconditional Legendre remains **OPEN**.

Abstract

Problem. Legendre: for every integer $n \geq 1$ there is a prime in $(n^2, (n+1)^2)$.

Literature (CONDITIONAL on RH). Chamberland–Straub (arXiv:2602.22502, 2026), combining Carneiro–Milinovich–Soundararajan short-interval primes under RH with Sorenson–Webster's Oppermann verification, obtain: if RH holds and $\delta \geq 1/4$, then $[x^{2+\delta}, (x+1)^{2+\delta}]$ contains a prime for all real $x \geq 1$. That is the **collapse** $\alpha = 2.25$ target. The same note indicates the threshold is sharpenable to $\delta \approx 0.2253$ ($\alpha \approx 2.2253$); $\delta = 1/5$ remains out of reach.

Literature (verified range). Sorenson–Webster (2025) verify **Oppermann's** conjecture (stronger than Legendre) for all $n \leq 7.05 \cdot 10^{13}$, hence Legendre holds at least to that n.

Box (method demo). On-box Eratosthenes + interval scan certifies Legendre for all $n = 1, \ldots, N$ with $N = 20{,}000$ (**0** failures; wall ≈ 14.9 s). This window is **not** the literature record.

Problem (Band-1 collapse item)

1. State Legendre honestly as **OPEN** (even under RH for squares).
2. Cite the **CONDITIONAL** RH collapse at $\alpha \geq 2.25$ (and zero-free cousins for higher powers).
3. Cite published verification ranges without inventing a record.
4. Re-check a feasible finite N on box; write `results.json`.
5. Keep Clay / ToE / GATE-04 off the claim surface.

Literature ledger

Unconditional status

Statement	Status
Prime in $(n^2, (n+1)^2)$ for all $n \geq 1$	**OPEN**

Statement	Status
Same under RH	Still **OPEN** for $\alpha = 2$
Almost-primes $\Omega(a) \leq 4$ in every such interval	Dudek–Johnston (JNT 2026) — unconditional almost-prime weakening

Verified ranges (not a proof)

Year	Result	Authors
2025	Oppermann (hence Legendre) for all $n \leq 7.05 \cdot 10^{13}$	Sorenson–Webster, *Res. Number Theory* **11**

CONDITIONAL results under RH (collapse ≈ 2.25)

Hypothesis	Conclusion	Authors / ref
RH	For $\delta \geq 1/4$ ($\alpha = 2 + \delta \geq 2.25$), $[x^\alpha, (x+1)^\alpha]$ has a prime for all $x \geq 1$	Chamberland–Straub, arXiv:2602.22502
RH	Same line sharpenable to $\delta \approx 0.2253$ ($\alpha \approx 2.2253$)	Chamberland–Straub §3 note
RH	For $x \geq 4$, prime in $[x, x + \frac{22}{25}\sqrt{x}\log x]$	Carneiro–Milinovich–Soundararajan, CMH **94** (2019)

Explicit honesty. RH + CMS19 short intervals give primes between consecutive α**-powers** for $\alpha > 2$ (sufficiently large x), and with SW25 for **all** $x \geq 1$ once $\alpha \geq 2.25$. They do **not** currently close $\alpha = 2$ (true Legendre), even conditionally on RH.

Zero-free region cousins (also CONDITIONAL)

Hypothesis	Conclusion	Ref
Stated minimal Littlewood-form zero-free region	Primes in $(n^k, (n+1)^k)$ for fixed $k \in \{70, 75, 80, 85, 90\}$	arXiv:2602.14340 (2026)
Unconditional milestone nearby	Primes between consecutive **86th** powers for all $n \geq 1$	related CHJ / 2602.14340 line

These weaken Legendre toward **higher** powers; they do not close squares.

Method (box)

1. **Sieve.** Eratosthenes boolean mask on $[0, (N+1)^2]$ with $N = 20{,}000$ (limit 400,040,001).
2. **Scan.** For each $n = 1, \ldots, N$, count primes in $(n^2, (n+1)^2)$; record first prime.
3. **Pass criterion.** Zero empty intervals.
4. **Label.** `BOX_METHOD_DEMO` — orders of magnitude below SW25; **not** a proof.

Results

CONDITIONAL status (literature)

Quantity	Value	Label
Collapse target	$\alpha \geq 2.25$ under **RH**	**CONDITIONAL**
Sharpening cited	$\alpha \approx 2.2253$	**CONDITIONAL**
True Legendre ($\alpha = 2$) under RH	**OPEN**	honest
Lit verified n (Oppermann)	$7.05 \cdot 10^{13}$	published

Box window (certified tonight)

Quantity	Value
N certified	20,000
Intervals checked	20,000
Failures	0
All verified	**True**
Min primes per interval	2
Max primes per interval	2,095
Mean primes per interval	≈ 1066.92
Sieve time	≈ 6.30 s
Scan time	≈ 8.58 s
Wall (box)	≈ 14.88 s

Sample first primes

n	$(n^2, (n+1)^2)$	first prime	#primes
1	$(1, 4)$	2	2
2	$(4, 9)$	5	2
3	$(9, 16)$	11	2
10	$(100, 121)$	101	5
100	$(10000, 10201)$	10007	23
1000	$(10^6, 1002001)$	1000003	152
10000	$(10^8, 100020001)$	100000007	1081
20000	$(4 \cdot 10^8, 400040001)$	400000009	2020

(Exact sample rows from `results.json`.)

Why Band-1 vs conjecture

- **Band-1 (this probe):** cite **CONDITIONAL** RH collapse at 2.25; cite published verification; certify finite N.
- **Conjecture (OPEN):** primes between **every** consecutive squares — **not proved here**, not even under RH.

Reproducibility

- Lab: `/workspace/mission-control/experiments/band1/legendre_conditional/`
- Run: `python3 legendre_verify.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `METHOD.md`

Conclusion

N **certified (box):** 20,000 (0 failures). **CONDITIONAL status:** under **RH**, primes between consecutive α-powers for all $x \geq 1$ when $\alpha \geq 2.25$ (Chamberland–Straub 2026; collapse target); true Legendre ($\alpha = 2$) remains **OPEN** even under RH. **Unconditional Legendre: OPEN**. No Clay claim. GATE-04 locked.